Long-Term Efficacy of Rituximab in Idiopathic Inflammatory Myopathies Refractory to csDMARDs - Preliminary Results of a Prospective Observational Study

Daniel Zavada[1,2], Sabina Hrabe Oreska[1,2], Anna Bobrova[1], Katerina Bobrova[1], Eliska Sudkova[1], Emma Polouckova[1], Martin Klein[1,2], Herman Mann[1,2], Ladislav Senolt[1,2], Jiri Vencovsky[1,2], Michal Tomcik[1,2]

[1]Institute of Rheumatology, Prague, Czech Republic; [2]Department of Rheumatology, First Faculty of Medicine, Charles University, Prague, Czech Republic

Introduction

- Evidence on the efficacy of bDMARDs in idiopathic inflammatory myopathies (IIM) non-responsive to glucocorticoids and csDMARDs is limited.
- Aim: to assess the long-term efficacy of rituximab (RTX) in bDMARD-naive IIM patients refractory to csDMARDs.

Methods

- 36 patients with IIM (fulfilling 2017 ACR/EULAR classification criteria) refractory to glucocorticoids and csDMARDs, treated with RTX 1 g i.v. on days 0 and 14 every 6 months. (Baseline characteristics in **Table 1**)
- Follow-up at month (M) 6 (n=30), M12 (n=23), M24 (n=19) and M36 (n=8).
- Before each RTX course:
  - myositis response (2016 ACR/EULAR Total Improvement Score, TIS)
  - disease activity (MITAX, MYOACT) and muscle strength (MMT-8)
  - laboratory parameters (CK, myoglobin, CRP, ESR) and glucocorticoid dose
  - pulmonary function (FVC, FEV1, DLCO)
  - PROs: function, quality of life, fatigue, depression, physical activity and respiratory symptoms (HAQ, SF-36, FIS, BDI-II, HAP, SGRQ)
- Modelled using linear mixed-effects models (random intercept per patient)

Table 1. Baseline characteristics (N = 36)

Median (IQR) or n (%) of patients with available data; glucocorticoid dose is reported among patients actively taking glucocorticoids (n=32)

Patient characteristics	
Age [years]	56 (45-66)
Female	25 (69%)
Disease duration [years]	6 (2-11)
ASyS	14 (39%)
IMNM	9 (25%)
DM	7 (19%)
DM + ASyS	4 (11%)
PM	2 (6%)
ILD	24 (67%)
MITAX	0.17 (0.15-0.26)
MYOACT	0.09 (0.03-0.13)
MMT-8	70 (63-76)
CK [µkat/L]	2.9 (1.2-29.8)
FVC [% pred.]	80 (59-95)
DLCO [% pred.]	67 (53-77)

Medication at baseline	
Glucocorticoids	32 (91%)
Prednisone eq. [mg/day]	10.0 (5.0-20.0)
Methotrexate	13 (36%)
Mycophenolate	11 (31%)
Azathioprine	1 (3%)
Cyclosporine A	2 (6%)
Tacrolimus	2 (6%)
Cyclophosphamide	1 (3%)

Figure 1. Myositis response and disease activity

Estimated means with 95% CI from linear mixed-effects models are shown in black, individual patient data in grey. n = number of patients assessed at that timepoint. Comparisons with M0: * p<0.05; ** p<0.01; *** p<0.001.

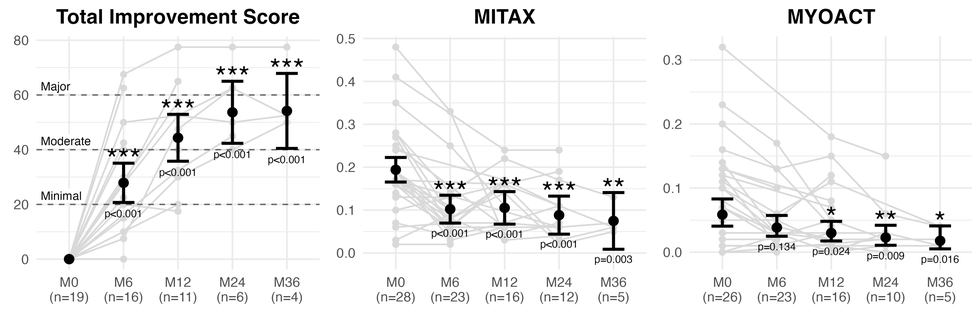
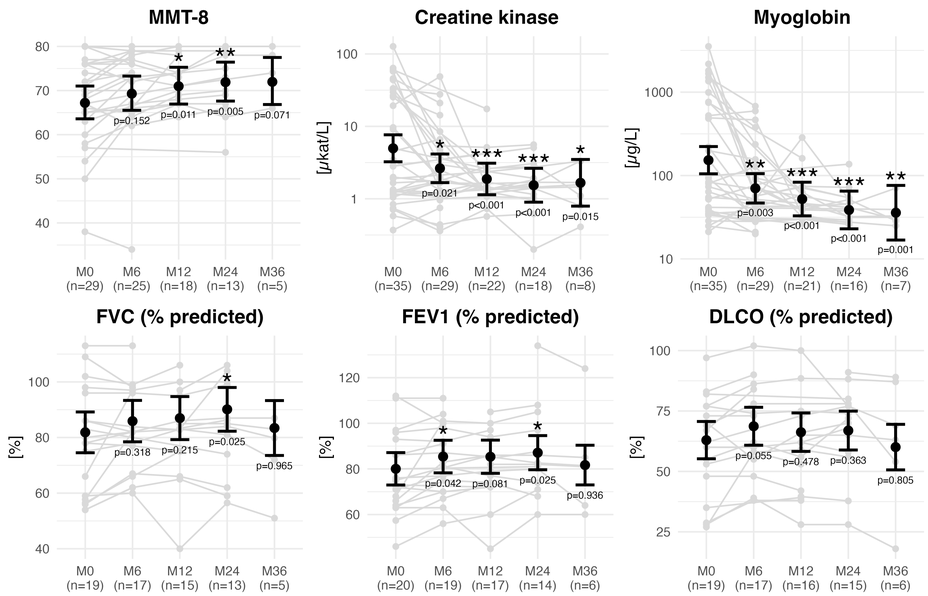


Figure 2. Muscle involvement and pulmonary function

Estimated means with 95% CI from linear mixed-effects models are shown in black, individual patient data in grey. n = number of patients assessed at that timepoint. Comparisons with M0: * p<0.05; ** p<0.01; *** p<0.001.

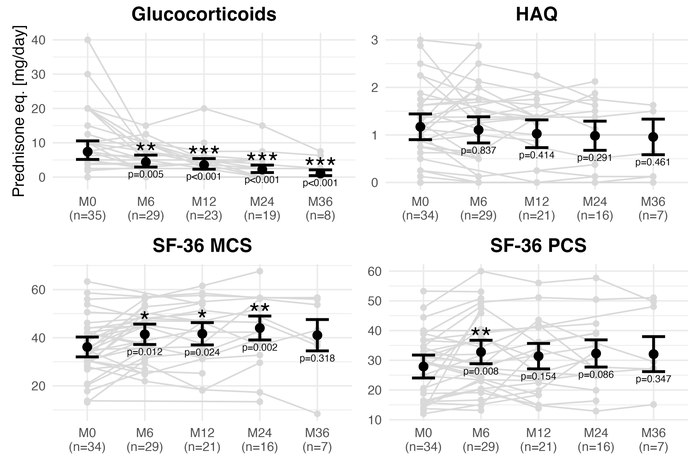


Results

- **Myositis response (Fig. 1)**: TIS 27.9 (M6), 44.4 (M12), 53.7 (M24) and 54.2 (M36; all p<0.001) - minimal improvement from M6, moderate from M12.
- **Disease activity and muscles (Fig. 1, 2)**: MITAX, CK and myoglobin decreased at all visits (CK 5.0 → 1.7 µkat/L, myoglobin 153 → 36 µg/L; M0 → M36), MYOACT from M12; MMT-8 improved at M12 and M24.
- **Pulmonary function (Fig. 2)**: FVC increased at M24 and FEV1 at M6 and M24; DLCO stable.
- **Glucocorticoids and patient-reported outcomes (Fig. 3)**: prednisone eq. dose tapered from 7.4 to 1.1 mg/day (M0 → M36; p<0.001); SF-36 MCS improved at M6, M12 and M24 and SF-36 PCS at M6.
- No significant change in HAQ, HAP, FIS, BDI-II, SGRQ, CRP or ESR.

Figure 3. Glucocorticoid dose and patient-reported outcomes

Higher SF-36 and lower HAQ scores = better
*Estimated means with 95% CI from linear mixed-effects models are shown in black (geometric means for the glucocorticoid dose), individual patient data in grey. n = number of patients assessed at that timepoint. Comparisons with M0: * p<0.05; ** p<0.01; *** p<0.001.*



Conclusions

- **Long-term RTX in csDMARD-refractory IIM was associated with a sustained clinical response and reduced disease activity and muscle enzymes up to M36.**
- **Muscle strength and mental health improved; glucocorticoids were tapered.**
- **Lung function was stable or slightly improved.**

Abbreviations: ASyS - antisynthetase syndrome; bDMARDs/csDMARDs - biologic/conventional synthetic disease-modifying antirheumatic drugs; BDI-II - Beck's Depression Inventory-II; CK - creatine kinase; CRP - C-reactive protein; DLCO - diffusing capacity of the lung for CO; DM - dermatomyositis (incl. clinically amyopathic DM); ESR - erythrocyte sedimentation rate; FEV1 - forced expiratory volume in 1 s; FIS - Fatigue Impact Scale; FVC - forced vital capacity; HAP - Human Activity Profile; HAQ - Health Assessment Questionnaire; IIM - idiopathic inflammatory myopathies; ILD - interstitial lung disease; IMNM - immune-mediated necrotizing myopathy; i.v. - intravenous; M - month; MCS/PCS - SF-36 mental/physical component summary; MITAX - Myositis Intention to Treat Activity Index; MMT-8 - Manual Muscle Testing; MYOACT - Myositis Disease Activity Assessment Visual Analogue Scales; PM - polymyositis; RTX - rituximab; SF-36 - Short Form-36 Health Survey; SGRQ - St George's Respiratory Questionnaire; TIS - Total Improvement Score.

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zavadad@revma.cz www.revma.cz

Supported by MHCR 023728, SVV 260764, LM2023033 - BBMRI.cz